

Model Development Phase Template

Date	28 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	4 Marks

Initial Model Training Code, Model Validation and Evaluation Report

The initial model training code will be showcased in the future through a screenshot. The model validation and evaluation report will include classification reports, accuracy, and confusion matrices for multiple models, presented through respective screenshots.

Initial Model Training Code:

Paste the screenshot of the model training code

Model Validation and Evaluation Report:

Model	Classification Report	Accuracy	Confusion Matrix
Random Forest	<pre>print(classification_report(y_test, y_test_predict1))</pre> <pre> precision recall f1-score support Fraud 0.97 0.97 0.97 221 is not Fraud 0.98 0.97 0.98 265 accuracy 0.97 0.97 0.97 486 macro avg 0.97 0.97 0.97 486 weighted avg 0.97 0.97 0.97 486 </pre>	0.9732510288065843	<pre>[139]: pd.crosstab(y_test,y_test_predict1)</pre> <pre> [139]: col_0 Fraud is not Fraud isFraud Fraud 215 6 is not Fraud 7 258 </pre>
Decision Tree	<pre>print(classification_report(y_test, y_test_predict2))</pre> <pre> precision recall f1-score support Fraud 0.92 0.94 0.93 221 is not Fraud 0.95 0.94 0.94 265 accuracy 0.94 0.94 0.94 486 macro avg 0.94 0.94 0.94 486 weighted avg 0.94 0.94 0.94 486 </pre>	0.9382716049382716	<pre>[143]: pd.crosstab(y_test,y_test_predict2)</pre> <pre> [143]: col_0 Fraud is not Fraud isFraud Fraud 208 13 is not Fraud 17 248 </pre>

Extra Trees Classifier	<pre>[146]: print(classification_report(y_test, y_test_predict3))</pre> <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>Fraud</td><td>0.97</td><td>0.95</td><td>0.96</td><td>221</td></tr><tr><td>is not Fraud</td><td>0.96</td><td>0.98</td><td>0.97</td><td>265</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.97</td><td>486</td></tr><tr><td>macro avg</td><td>0.97</td><td>0.96</td><td>0.96</td><td>486</td></tr><tr><td>weighted avg</td><td>0.97</td><td>0.97</td><td>0.96</td><td>486</td></tr></tbody></table>		precision	recall	f1-score	support	Fraud	0.97	0.95	0.96	221	is not Fraud	0.96	0.98	0.97	265	accuracy			0.97	486	macro avg	0.97	0.96	0.96	486	weighted avg	0.97	0.97	0.96	486	0.9650	<pre>[147]: pd.crosstab(y_test,y_test_predict3)</pre> <pre>[147]:</pre> <table><thead><tr><th></th><th>col_0</th><th>Fraud</th><th>is not Fraud</th></tr></thead><tbody><tr><td>isFraud</td><td></td><td></td><td></td></tr><tr><td>Fraud</td><td></td><td>210</td><td>11</td></tr><tr><td>is not Fraud</td><td></td><td>6</td><td>259</td></tr></tbody></table>		col_0	Fraud	is not Fraud	isFraud				Fraud		210	11	is not Fraud		6	259					
	precision	recall	f1-score	support																																																		
Fraud	0.97	0.95	0.96	221																																																		
is not Fraud	0.96	0.98	0.97	265																																																		
accuracy			0.97	486																																																		
macro avg	0.97	0.96	0.96	486																																																		
weighted avg	0.97	0.97	0.96	486																																																		
	col_0	Fraud	is not Fraud																																																			
isFraud																																																						
Fraud		210	11																																																			
is not Fraud		6	259																																																			
Support Vector Classification	<pre>1: print(pd.crosstab(y_test,y_test_predict4)) print('\n') print(classification_report(y_test, y_test_predict4))</pre> <table><thead><tr><th>col_0</th><th>Fraud</th><th>is not Fraud</th></tr></thead><tbody><tr><td>isFraud</td><td>111</td><td>110</td></tr><tr><td>is not Fraud</td><td>2</td><td>263</td></tr></tbody></table> <table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>Fraud</td><td>0.98</td><td>0.50</td><td>0.66</td><td>221</td></tr><tr><td>is not Fraud</td><td>0.71</td><td>0.99</td><td>0.82</td><td>265</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.77</td><td>486</td></tr><tr><td>macro avg</td><td>0.84</td><td>0.75</td><td>0.74</td><td>486</td></tr><tr><td>weighted avg</td><td>0.83</td><td>0.77</td><td>0.75</td><td>486</td></tr></tbody></table>	col_0	Fraud	is not Fraud	isFraud	111	110	is not Fraud	2	263		precision	recall	f1-score	support	Fraud	0.98	0.50	0.66	221	is not Fraud	0.71	0.99	0.82	265	accuracy			0.77	486	macro avg	0.84	0.75	0.74	486	weighted avg	0.83	0.77	0.75	486	0.7695473251028807	<pre>print(pd.crosstab(y_test,y_test_predict4)) print('\n') print(classification_report(y_test, y_test_predict4))</pre> <table><thead><tr><th>col_0</th><th>Fraud</th><th>is not Fraud</th></tr></thead><tbody><tr><td>isFraud</td><td></td><td></td></tr><tr><td>Fraud</td><td>111</td><td>110</td></tr><tr><td>is not Fraud</td><td>2</td><td>263</td></tr></tbody></table>	col_0	Fraud	is not Fraud	isFraud			Fraud	111	110	is not Fraud	2	263
col_0	Fraud	is not Fraud																																																				
isFraud	111	110																																																				
is not Fraud	2	263																																																				
	precision	recall	f1-score	support																																																		
Fraud	0.98	0.50	0.66	221																																																		
is not Fraud	0.71	0.99	0.82	265																																																		
accuracy			0.77	486																																																		
macro avg	0.84	0.75	0.74	486																																																		
weighted avg	0.83	0.77	0.75	486																																																		
col_0	Fraud	is not Fraud																																																				
isFraud																																																						
Fraud	111	110																																																				
is not Fraud	2	263																																																				
XG boost	<table><thead><tr><th></th><th>precision</th><th>recall</th><th>f1-score</th><th>support</th></tr></thead><tbody><tr><td>0</td><td>0.98</td><td>0.98</td><td>0.98</td><td>221</td></tr><tr><td>1</td><td>0.98</td><td>0.98</td><td>0.98</td><td>265</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.98</td><td>486</td></tr><tr><td>macro avg</td><td>0.98</td><td>0.98</td><td>0.98</td><td>486</td></tr><tr><td>weighted avg</td><td>0.98</td><td>0.98</td><td>0.98</td><td>486</td></tr></tbody></table>		precision	recall	f1-score	support	0	0.98	0.98	0.98	221	1	0.98	0.98	0.98	265	accuracy			0.98	486	macro avg	0.98	0.98	0.98	486	weighted avg	0.98	0.98	0.98	486	0.9835390946502057	<pre>print(pd.crosstab(y_test1,y_test_predict5)) print('\n') print(classification_report(y_test1, y_test_predict5))</pre> <table><thead><tr><th>col_0</th><th>0</th><th>1</th></tr></thead><tbody><tr><td>row_0</td><td></td><td></td></tr><tr><td>0</td><td>217</td><td>4</td></tr><tr><td>1</td><td>4</td><td>261</td></tr></tbody></table>	col_0	0	1	row_0			0	217	4	1	4	261									
	precision	recall	f1-score	support																																																		
0	0.98	0.98	0.98	221																																																		
1	0.98	0.98	0.98	265																																																		
accuracy			0.98	486																																																		
macro avg	0.98	0.98	0.98	486																																																		
weighted avg	0.98	0.98	0.98	486																																																		
col_0	0	1																																																				
row_0																																																						
0	217	4																																																				
1	4	261																																																				